

Reverse Turing Test

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CS375

1. Introduction

This Reverse Turing Test is a social deduction game that was programmed using python and Streamlit. You are a human put against 4 AI players. Each round everyone will be asked a question, which you (the human) must try to respond like an AI to avoid being voted out. After everyone answers, the AI players will vote out who they think has the most human response, while you must vote which AI you believe they'd vote for to lean the vote in your favor. The game is designed to be not only entertaining but educational as well. You will learn subtle patterns and differences between human and AI writing.

2. Accessing The Game

There is no download required for the game, as it is hosted on the Streamlit community cloud.

You do not need an account to play the game. To access:

- Open the game URL in your browser
- Read the rules on the welcome screen
- Click the "Start Game" button
- Wait briefly for the game to set up the players and generate the rounds
- After the game is generated, you will see your randomly assigned name at the top of the screen

3. How to play

3.1. Submitting a Response

At the start of each round, a random question will be prompted to all players and displayed at the top of the screen. All four AI players' names, responses, and randomly assigned personalities will be displayed automatically and displayed on screen. Your job is to read through all AI responses and write your own response in the textbox near the bottom of the screen. Try to sound like an AI by copying their writing styles to stay in the game. Then press “Submit Response” once you have finished typing your answer. (Your answer must fit within a 255 character limit).

3.2. Voting Phase

After you submit your answer, the Voting Phase will begin. You will see your response displayed along with the list of the active AI players in the game. You (the human player) will have a checkbox displaying the AI player’s names, in which you can select who you want to vote out of the next round. The AI players will also be casting their own vote for who they think is the human. After you select the player you want voted out, you click the “Submit Vote” button.

3.3. Vote Results

After all votes are counted, the results are displayed on the screen showing how many votes each player received. The player with the most votes against them is eliminated each round.

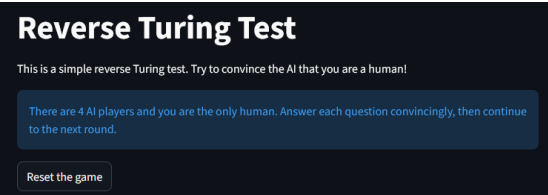
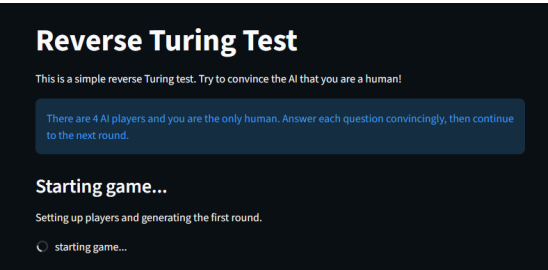
Outcome	What Happens
An AI is eliminated	You survive. Click the “Next Round” button to continue

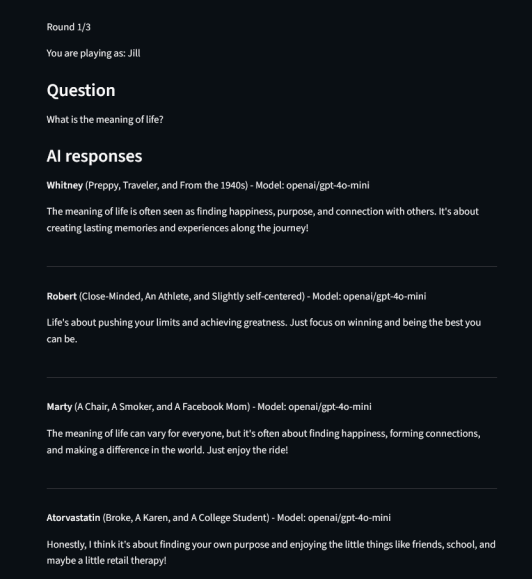
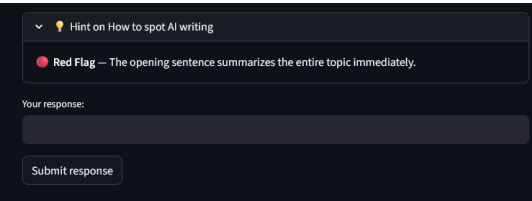
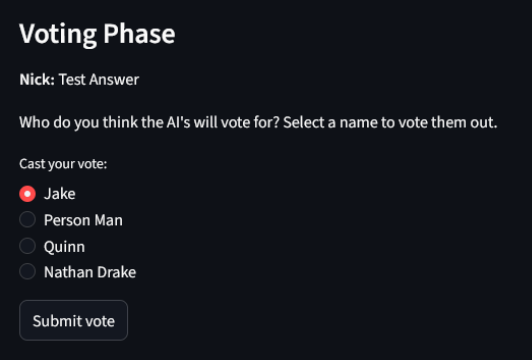
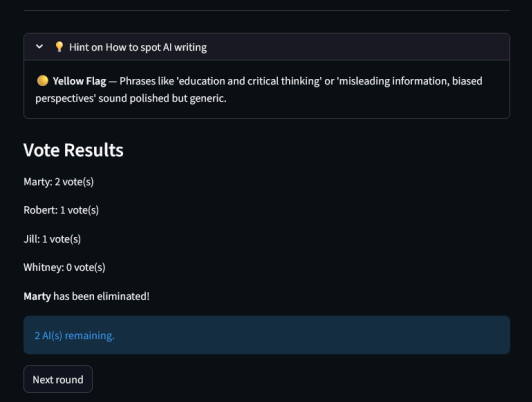
All players get 1 vote each	A random AI is eliminated. Click the “Next Round” button to continue
You are eliminated	Game Over. The AIs win. Click “Play again” to restart.
After the final round if you are not voted out	You survived all rounds. You win! Click “Play again” to restart the game.

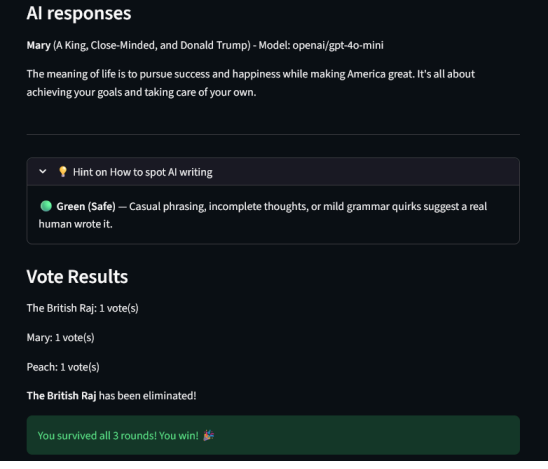
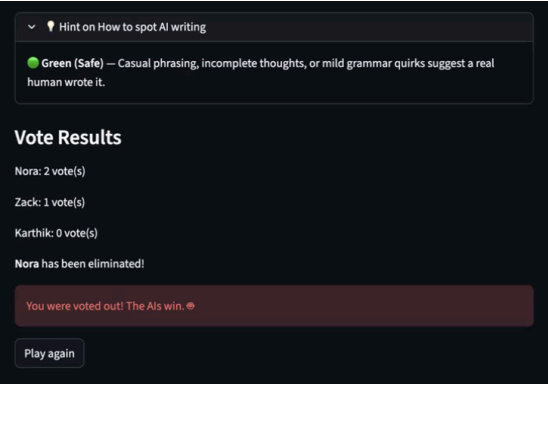
3.4 Winning & Losing

The game lasts up to 3 rounds. One AI is eliminated each round. If you survive all 3 rounds without being voted out, you win. If you receive the most votes in any round, you are eliminated and the AIs win the game.

4. Game Screens Reference (Screenshots of each screen here)

Screen	Description
	Displays the rules and the “Start Game” button
	Briefly Shown while the game generates the round and the player names.

 Round 1/3 You are playing as: Jill Question What is the meaning of life? AI responses Whitney (Preppy, Traveler, and From the 1940s) - Model: openai/gpt-4o-mini The meaning of life is often seen as finding happiness, purpose, and connection with others. It's about creating lasting memories and experiences along the journey! Robert (Close-Minded, An Athlete, and Slightly self-centered) - Model: openai/gpt-4o-mini Life's about pushing your limits and achieving greatness. Just focus on winning and being the best you can be. Marty (A Chair, A Smoker, and A Facebook Mom) - Model: openai/gpt-4o-mini The meaning of life can vary for everyone, but it's often about finding happiness, forming connections, and making a difference in the world. Just enjoy the ride! Atorvastatin (Broke, A Karen, and A College Student) - Model: openai/gpt-4o-mini Honestly, I think it's about finding your own purpose and enjoying the little things like friends, school, and maybe a little retail therapy!	Shows the round number, your player name, the question, and all of the AI responses. Hints will also be shown towards the bottom of the screen.
 Hint on How to spot AI writing Red Flag — The opening sentence summarizes the entire topic immediately. Your response: Submit response	The text box and submit button at the bottom of the round screen. Appears after you read through all AI responses towards the bottom of the screen
 Voting Phase Nick: Test Answer Who do you think the AI's will vote for? Select a name to vote them out. Cast your vote: ☒ Jake ☐ Person Man ☐ Quinn ☐ Nathan Drake Submit vote	Appears after you submit your answer. Shows a checkbox list of active AI player names to vote on.
 Hint on How to spot AI writing Yellow Flag — Phrases like 'education and critical thinking' or 'misleading information, biased perspectives' sound polished but generic. Vote Results Marty: 2 vote(s) Robert: 1 vote(s) Jill: 1 vote(s) Whitney: 0 vote(s) Marty has been eliminated! 2 AI(s) remaining. Next round	Shows the vote counts and who was eliminated from the game. You can either continue or restart depending on whether or not you were voted out.

	Win message displayed with the “Play Again” button at the bottom.
	Loss message displayed with the “Play Again” button at the bottom.

5. Reading Hints

During the first three rounds of the game, a collapsible hint panel appears below the AI responses. Each hint teaches you a pattern to watch for when reading AI generated text. The hints change by category each round:

Round	Hint Type
Round 1	 Red Flag: Strong AI signals. Human writers generally do not do this.
Round 2	 Yellow Flag: Mild AI Signals. Humans

	can write this way, but AI typically does it more often
Round 3	 Green (Safe): Could pass as a human written response

6. FAQ

Why does the game sometimes take a while to start?

The game calls a live AI API to generate responses for all four AI players. The network request can take anywhere between 5-15 seconds depending on server load.

Why is the text box limited to 255 characters?

This matches the same character limit as the AI players, keeping the game fair. It encourages concise answers and prevents the AI from hallucinating.

Can I get the same name as an AI player?

No, you cannot. The system is set up so that every player, AI and human, have unique names.

Not only does this differentiate the AI players for the human, it prevents the AI players, who vote by declaring a name, to accidentally submit 2 votes.

What happens if there is a tie in the vote?

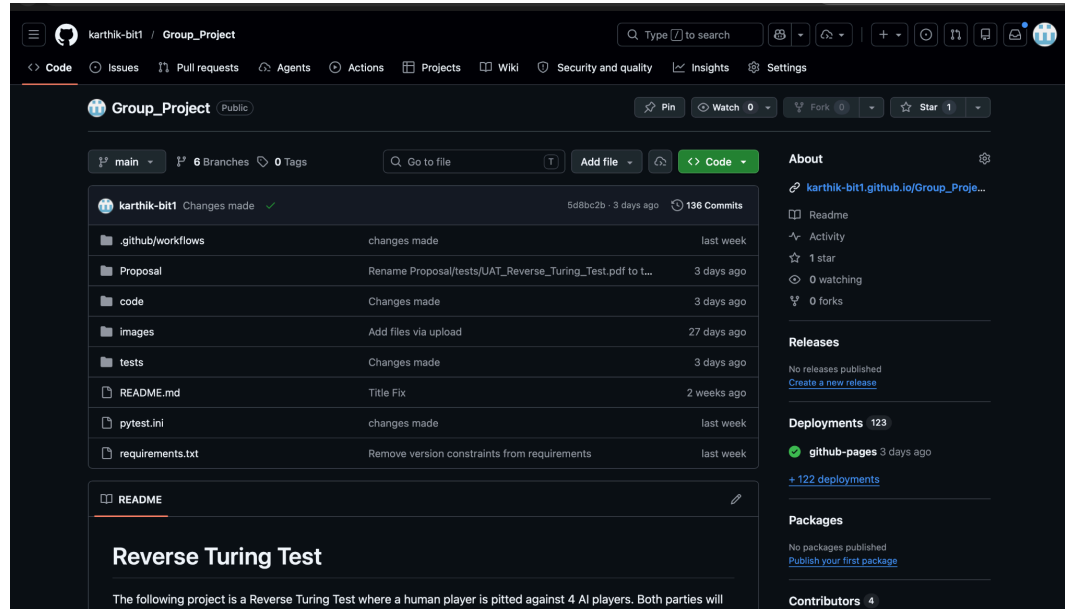
Can I reset mid-game?

Yes. The “Reset the Game” button is available at the top of every round screen. Pressing it immediately ends the current game and returns you to the welcome screen.

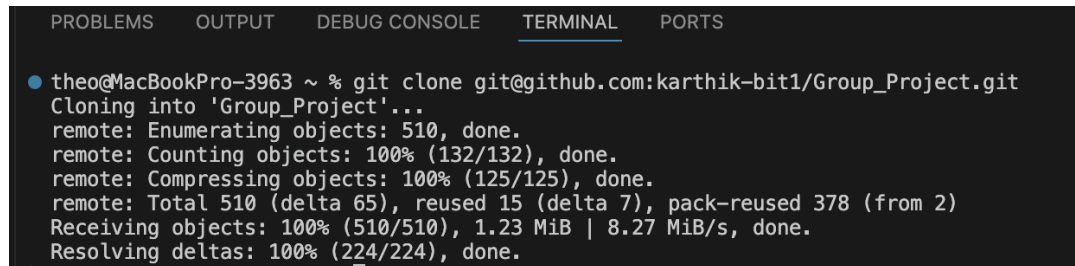
7. How to run the website on local computer

Go to the github repository: https://github.com/karthik-bit1/Group_Project

1. Fork the repository.



2. `git clone git@github.com:karthik-bit1/Group\_Project.git`



3. Set your GITHUB_TOKEN in a .env file or as an environment variable.

```
echo "GITHUB_TOKEN=your_personal_access_token_here" >> .env
```

(mac/linux, for windows/ git bash don't put quotes)



4. `python/python3 -m venv .venv`



5. `source .venv/bin/activate` (mac/ linux) `## source .venv/Scripts/activate` (git bash)

```
theo@MacBookPro-3963 ~ % source .venv/bin/activate
(.venv) theo@MacBookPro-3963 ~ %
```

6. `cd Group_Project`

```
(.venv) theo@MacBookPro-3963 ~ % cd Group_Project
(.venv) theo@MacBookPro-3963 Group_Project %
```

7. `pip/pip3 install -r requirements.txt`

```
(.venv) theo@MacBookPro-3963 Group_Project % pip3 install -r requirements.txt
Using cached pandas-3.0.2-cp314-cp314-macosx_11_0_arm64.whl (9.9 MB)
Using cached pillow-11.2.0-cp314-cp314-macosx_11_0_arm64.whl (4.7 MB)
Using cached protobuf-7.34.1-cp318-abi3-macosx_10_9_universal2.whl (429 kB)
Using cached pydeck-0.9.2-py3-none-any.whl (11.3 MB)
Using cached requests-2.31.1-py3-none-any.whl (164 kB)
Using cached charset-normalizer-3.4.0-cp314-cp314-macosx_10_15_universal2.whl (309 kB)
Using cached idna-3.13-py3-none-any.whl (68 kB)
Using cached smmap-5.0.2-py3-none-any.whl (24 kB)
Using cached tenacity-9.1.4-py3-none-any.whl (28 kB)
Using cached tomli-2.0.1-py3-none-any.whl (16 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Using cached urllib3-2.5.0-py3-none-any.whl (121 kB)
Using cached azure_ai_inference-1.0.0b9-py3-none-any.whl (124 kB)
Using cached azure_core-1.39.0-py3-none-any.whl (124 kB)
Using cached python_dotenv-1.2.2-py3-none-any.whl (22 kB)
Using cached pytest-8.0.3-py3-none-any.whl (376 kB)
Using cached pluggy-1.6.0-py3-none-any.whl (20 kB)
Using cached pytest_cov-7.0.0-py3-none-any.whl (22 kB)
Using cached pytest_mock-3.15.1-py3-none-any.whl (10 kB)
Using cached anyio-4.13.0-py3-none-any.whl (114 kB)
Using cached certifi-2026.4.22-py3-none-any.whl (135 kB)
Using cached coverage-7.11.5-cp314-cp314-macosx_11_0_arm64.whl (210 kB)
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Using cached jsonschema-specifications-2025.9.1-py3-none-any.whl (12 kB)
Using cached MarkupSafe-3.0.3-cp314-cp314-macosx_11_0_arm64.whl (12 kB)
Using cached markupsafe-3.0.3-cp314-cp314-macosx_11_0_arm64.whl (12 kB)
Using cached packaging-26.2-py3-none-any.whl (180 kB)
Using cached pyarrow-24.0.0-cp314-cp314-macosx_11_0_arm64.whl (35.0 MB)
Using cached pygments-2.28.0-py3-none-any.whl (1.2 MB)
Using cached python_dateutil-2.9.0-py3-none-any.whl (229 kB)
Using cached python_multipart-0.0.27-py3-none-any.whl (29 kB)
Using cached rdpd-0.30.0-cp314-cp314-macosx_11_0_arm64.whl (353 kB)
Using cached s1c-1.17.0-py3-none-any.whl (11 kB)
Using cached starlette-1.0.0-py3-none-any.whl (72 kB)
Using cached uvicorn-0.46.0-py3-none-any.whl (74 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached websockets-16.0-cp314-cp314-macosx_11_0_arm64.whl (175 kB)
Installing collected packages: websockets, urllib3, typing_extensions, tomli, tenacity, smmap, six, rdpd-py, python-multipart, python-dotenv, pygments, pyarrow, protobuf, pluggy, pillow, packaging, numpy, nornalys, MarkupSafe, itsdangerous, isodate, httpcore, httpx, idna, httptools, h11, coverage, click, charset-normalizer, certifi, cachetools, blinker, attrs, uvicorn, requests, referencing, python-dateutil, pytest, Jinja2, gitdb, anyio, starlette, pytest-mock, pytest-cov, pydeck, pandas, jsonschema-specifications, gitpython, azure-core, jsonschema, azure-ai-inference, altair, strea
Successfully installed MarkupSafe-3.0.3 altair-6.1.0 anyio-4.13.0 attrs-28.1.0 azure-ai-inference-1.0.0b9 azure-core-1.39.0 blinker-1.9.0 cachetools-7.0.0 certifi-2026.4.22 charset-normalizer-3.4.0 click-8.3.1 coverage-7.11.5 gitdb-4.0.32 gitpython-3.1.40 h11-0.16.0 httpcore-2.1.0 httpx-0.7.1 idna-3.13 iniconfig-2.3.0 isodate-0.7.2 itsdangerous-2.2.0 Jinja2-3.1.6 jsonschema-4.26.0 jsonschema-specifications-2025.9.1 MarkupSafe-3.0.3 numpy-2.4.4 packaging-26.2 pandas-3.0.2 pillow-12.2.0 pluggy-1.6.0 protobuf-7.34.1 pyarrow-24.0.0 pydeck-0.9.2 pygments-2.28.0 pytest-8.0.3 pytest-cov-7.1.0 p
test-mock-3.15.1 python-dateutil-2.9.0 python-dotenv-1.2.2 python-multipart-0.0.27 referencing-0.37.0 requests-2.31.1 rdpd-py-0.30.0 six-1.17.0 smmap-5.0.2 starlette-1.0.0 streamlit-1.37.
tenacity-9.1.4 tomli-2.0.1 typing_extensions-4.15.0 urllib3-2.5.0 uvicorn-0.46.0 websockets-16.0
```

8. `cd code`

```
(.venv) theo@MacBookPro-3963 Group_Project % cd code
(.venv) theo@MacBookPro-3963 code %
```

9. `STREAMLIT_SERVER_ADDRESS=0.0.0.0`

`STREAMLIT_SERVER_PORT=8501 streamlit run reverse_turing_test.py`

```
(.venv) theo@MacBookPro-3963 code % STREAMLIT_SERVER_ADDRESS=0.0.0.0 STREAMLIT_SERVER_PORT=8501 streamlit run reverse_turing_test.py
2026-04-30 18:05:48.389 Uvicorn server started on 0.0.0.0:8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.0.116:8501

For better performance, install the Watchdog module:

$ xcode-select --install
$ pip install watchdog
```

8. System Architecture

The Reverse Turing Test is a single file Python application that runs with Streamlit. All game state is managed through Streamlit's session state directory. The AI responses are generated with Azure AI Inference API.

8.1 Components

Component	Role
Browser (User)	Renders the Streamlit UI. Sends user inputs back to the server for each interaction
Streamlit Server	Runs the Python application. Handles reruns triggered by button clicks and input changes
st.session_state	Persists all game data across reruns within a single browser session
Azure AI Inference API	Receives prompts and returns AI generated text. Used for both answer generation and AI voting
openai/gpt-4o-mini	Primary model used for AI players 1, 2, 3 and 4

8.3 Data Flow

- User clicks “Start Game” —> name, personalities sampled and stored in session_state
- generate_round() increments round counter, picks a question, calls responseAI()
- responseAI() send 4 prompts to the API and stores the responses in ai_answers()
- User reads answers, types their response clicks “Submit Response” —> phase set to “voting”
- callvote() renders voting list, user selects a name and clicks “Submit Vote”

- `tally_votes()` counts human vote and calls `ai_vote()` once for each active AI
- Each `ai_vote()` call sends all answers to the AI and asks it to identify the human
- Most voted player is added to the eliminated list —> set phase to “result”
- `show_result()` checks win/lose conditions and either ends the game or advances the round